

MH1812 Tutorial

Chapter 6: Linear Recurrence

Q1: Solve the recurrence relation $a_n = 2a_{n-1} - a_{n-2}$ with initial values $a_1 = 3, a_0 = 0$ using

- (a) the backtracking method.
- (b) the characteristic equation.

Solution:

- (a) We have $a_n = 2a_{n-1} - a_{n-2}$, thus $a_{n-1} = 2a_{n-2} - a_{n-3}$, $a_{n-2} = 2a_{n-3} - a_{n-4}$, $a_{n-3} = 2a_{n-4} - a_{n-5}$, and so on. Therefore

$$\begin{aligned}a_n &= 2a_{n-1} - a_{n-2} \\&= 2(2a_{n-2} - a_{n-3}) - a_{n-2} = 3a_{n-2} - 2a_{n-3} \\&= 3(2a_{n-3} - a_{n-4}) - 2a_{n-3} = 4a_{n-3} - 3a_{n-4} \\&= 4(2a_{n-4} - a_{n-5}) - 3a_{n-4} = 5a_{n-4} - 4a_{n-5} \\&= \dots\end{aligned}$$

We see that a general term is $(i+1)a_{n-i} - ia_{n-(i+1)}$. Therefore the last term is when $n-i-1=0$ that is $i=n-1$, for which we have $na_1 - (n-1)a_0$, therefore with initial condition $a_0 = 0$ and $a_1 = 3$, we get

$$a_n = 3n.$$

- (b) The characteristic equation is $x^2 - 2x + 1$ that is

$$(x-1)^2 = 0,$$

which has a multiple root $x_1 = x_2 = 1$. Therefore

$$a_n = c_1 \cdot 1^n + c_2 \cdot n \cdot 1^n = c_1 + c_2 n.$$

The initial conditions are:

$$\begin{aligned}a_0 &= c_1 = 0 \\a_1 &= c_1 + c_2 = 3.\end{aligned}$$

Solving above two equations, we obtain:

$$c_1 = 0, \quad c_2 = 3.$$

Hence, $a_n = 3n$. □

Q2: Solve the recurrence relation

$$a_n = 2a_{n-1} + a_{n-2} - 2a_{n-3}$$

with initial values $a_0 = 1$, $a_1 = 2$ and $a_2 = 0$.

Solution: The characteristic equation of this recurrence is:

$$x^3 - 2x^2 - x + 2 = 0,$$

or,

$$x^2(x-2) - (x-2) = 0 \Leftrightarrow (x^2-1)(x-2) = 0 \Leftrightarrow (x-1)(x+1)(x-2) = 0.$$

It has three distinct roots $x_1 = 1$, $x_2 = -1$, $x_3 = 2$. Therefore

$$a_n = c_1 \cdot 1^n + c_2 \cdot (-1)^n + c_3 \cdot 2^n = c_1 + c_2(-1)^n + c_3 \cdot 2^n$$

with initial values

$$a_0 = c_1 + c_2 + c_3 = 1,$$

$$a_1 = c_1 - c_2 + 2c_3 = 2,$$

$$a_2 = c_1 + c_2 + 4c_3 = 0.$$

Solving above results in $c_1 = 2$, $c_2 = -2/3$ and $c_3 = -1/3$. Therefore

$$a_n = 2 - \frac{2}{3}(-1)^n - \frac{1}{3}2^n. \quad \square$$

Q3: Let $a_n = c_1a_{n-1} + c_2a_{n-2} + \dots + c_ka_{n-k}$ be a linear homogeneous recurrence. Assume both sequences a_n, a'_n satisfy this linear homogeneous recurrence. Show that $a_n + a'_n$ and αa_n also satisfy it, for α some constant.

Solution:

$$\begin{aligned} a_n + a'_n &= (c_1a_{n-1} + c_2a_{n-2} + \dots + c_ka_{n-k}) + (c_1a'_{n-1} + c_2a'_{n-2} + \dots + c_ka'_{n-k}) \\ &= c_1(a_{n-1} + a'_{n-1}) + c_2(a_{n-2} + a'_{n-2}) + \dots + c_k(a_{n-k} + a'_{n-k}) \end{aligned}$$

Thus $a_n + a'_n$ is a solution of the recurrence. Similarly,

$$\begin{aligned} \alpha a_n &= \alpha(c_1a_{n-1} + c_2a_{n-2} + \dots + c_ka_{n-k}) \\ &= c_1\alpha a_{n-1} + c_2\alpha a_{n-2} + \dots + c_k\alpha a_{n-k} \end{aligned}$$

Therefore αa_n is a solution of the recurrence. $\square$

Q4: Solve the recurrence relation

$$a_n = 4a_{n-1} - 3a_{n-2}$$

with the initial values $a_1 = 0$ and $a_2 = 12$.

Solution: The characteristic equation is $x^2 = 4x - 3$, or, $x^2 - 4x + 3 = (x - 3)(x - 1) = 0$, which has two distinct roots $x_1 = 3$ and $x_2 = 1$. Therefore $a_n = c_1 \cdot 3^n + c_2 \cdot 1^n = c_1 \cdot 3^n + c_2$. The initial conditions are:

$$a_1 = 3c_1 + c_2 = 0$$

$$a_2 = 9c_1 + c_2 = 12$$

Solving above results in $c_1 = 2$, and $c_2 = -6$, Hence $a_n = 2 \cdot 3^n - 6$. □

Q5: Solve the recurrence relation

$$a_n = 3a_{n-1} - 1$$

with initial value $a_1 = 1$.

Solution: We can use backtracking to solve the recurrence relation.

$$\begin{aligned} a_n &= 3a_{n-1} - 1 \\ &= 3(3a_{n-2} - 1) - 1 \\ &= 3^2 a_{n-2} - (3 + 1) \\ &= 3^2 (3a_{n-3} - 2) - (3 + 1) \\ &= 3^3 a_{n-3} - (3^2 + 3 + 1) \\ &= \dots \\ &= 3^{n-1} a_1 - (3^{n-2} + \dots + 3^2 + 3 + 1) \\ &= 3^{n-1} - \frac{3^{n-1} - 1}{2} \\ &= \frac{3^{n-1} + 1}{2}. \end{aligned}$$

Alternatively, we can use the characteristic equation. To this end, we observe that

$$a_n = 3a_{n-1} - 1$$

$$a_{n-1} = 3a_{n-2} - 1$$

Subtract the second equation from the first equation,

$$a_n - a_{n-1} = 3a_{n-1} - 3a_{n-2},$$

or,

$$a_n - 4a_{n-1} + 3a_{n-2} = 0.$$

The corresponding characteristic equation is

$$x^2 - 4x + 3 = 0,$$

which has two distinct roots $x_1 = 3$ and $x_2 = 1$. Therefore

$$a_n = c_1 \cdot 3^n + c_2 \cdot 1^n = c_1 \cdot 3^n + c_2.$$

Plugging in initial values $a_1 = 1$ and $a_2 = 2$,

$$3c_1 + c_2 = 1,$$

$$9c_1 + c_2 = 2,$$

whence we can solve that $c_1 = 1/6$ and $c_2 = 1/2$. Therefore

$$a_n = \frac{1}{6} \cdot 3^n + \frac{1}{2} = \frac{3^{n-1} + 1}{2}. \quad \square$$

Q6: Solve the recurrence relation:

$$a_n = 4a_{n-1} - 4a_{n-2}$$

with initial values $a_0 = 1$, $a_1 = 3$.

Solution: The characteristic equation is $x^2 = 4x - 4$, or, $x^2 - 4x + 4 = (x - 2)^2 = 0$, which has a multiple root $x_1 = x_2 = 2$. Therefore $a_n = c_1 \cdot 2^n + c_2 \cdot n2^n$. The initial conditions are:

$$a_0 = c_1 = 1,$$

$$a_1 = 2c_1 + 2c_2 = 3.$$

Solving above results in $c_1 = 1$ and $c_2 = 1/2$. Hence

$$a_n = 2^n + \frac{1}{2}n2^n = (n + 2)2^{n-1}. \quad \square$$